WORKSHEET — UNIT IV

Data Preprocessing and Visualization

Course: Data Science with Python **Unit:** IV (12 Hrs)

Name: _____ **Roll No:** _____ **Date:** _____

Topics covered: Loading data • Merge & reshape • Groupby • Pivot tables • Matplotlib & Seaborn

Section 1 — Key Concepts (Quick Revision)

Term	Simple Meaning	Example
CSV / file formats	CSV is a simple text file for table data. Pandas also reads Excel, JSON and SQL.	`pd.read_csv("data.csv")`
Data wrangling	Cleaning and preparing raw data for analysis.	`drop duplicates, fix NaN`
merge()	Joins two tables using a common column.	`pd.merge(a, b, on="id")`
concat()	Stacks tables one below the other.	`pd.concat([a, b])`
Reshaping	Changing data between wide and long layout.	`df.melt()  df.pivot()`
String manipulation	Cleaning the text in a column.	`df["name"].str.upper()`
groupby()	Split into groups, apply a function, combine the result.	`df.groupby("city")["sales"].sum()`
Pivot table	Summary table of values across two categories.	`df.pivot_table(...)`
Crosstab	Table of counts between two columns.	`pd.crosstab(df.a, df.b)`
Matplotlib	Basic Python library for drawing charts.	`plt.plot(x, y)`
Seaborn	Built on matplotlib. Gives nicer statistical charts.	`sns.barplot(x=..., y=...)`
Time series	Data in time order. A line chart shows the trend.	`plt.plot(month, sales)`

Section 2 — Multiple-Choice Questions

Choose the correct option and circle the letter.

1. **Which Pandas function reads data from a CSV file?**
   - **A.** pd.load_data() **B.** pd.read_csv()
   - **C.** pd.open_csv() **D.** pd.import_csv()

2. **Which function joins two DataFrames using a common column?**
   - **A.** concat() **B.** pivot()
   - **C.** merge() **D.** melt()

3. **Which method splits data into groups before applying a function?**
   - **A.** merge() **B.** read_csv()
   - **C.** plot() **D.** groupby()

4. **Changing data between wide and long layout is called:**
   - **A.** Reshaping **B.** Merging
   - **C.** Sorting **D.** Deleting rows

5. **Which table summarizes values across two categories?**
   A. Pivot table **B.** Series
   C. ndarray **D.** Tuple

6. **Which library is built on matplotlib and gives nicer statistical charts?**
   A. NumPy **B.** SQLAlchemy
   C. Seaborn **D.** BeautifulSoup

7. **Which matplotlib function draws a line chart?**
   A. plt.bar() **B.** plt.plot()
   C. plt.pie() **D.** plt.hist()

8. **Data points that are ordered by time are called:**
   A. Cross tabulation **B.** String manipulation
   C. Data merging **D.** Time series

9. **Which function makes a count (frequency) table of two columns?**
   A. pd.merge() **B.** pd.crosstab()
   C. pd.pivot() **D.** pd.concat()

10. **Which of these is a data wrangling task?**
    A. Compiling machine code **B.** Drawing a GUI window
    C. Fixing missing or duplicate values **D.** Sending an HTTP request

Section 3 — Fill in the Blanks

Word bank: merge() pivot groupby Seaborn time series

1. The _____ function joins two DataFrames using a shared column.
2. A _____ table summarizes data across two categories.
3. The _____ operation splits data into groups, applies a function, and combines the results.
4. _____ is a library built on matplotlib for statistical charts.
5. Data that is ordered by time is called _____ data.

Section 4 — True or False

1. merge() joins two DataFrames using a common column. **(True / False)**
2. A pivot table can only use the sum function. **(True / False)**
3. Seaborn is built on top of matplotlib. **(True / False)**
4. concat() joins tables using a common key column. **(True / False)**
5. A crosstab shows counts between two columns. **(True / False)**

Section 5 — Match the Following

Write the correct letter from Column B next to each item in Column A.

No.	Column A	Ans.	Column B
1	merge()		a. Basic library for drawing charts
2	groupby()		b. Joins tables on a common column
3	Pivot table		c. Data ordered by time
4	Seaborn		d. Splits data into groups
5	Matplotlib		e. Gives nicer statistical charts
6	Time series		f. Summarizes data across two categories

Section 6 — Short-Answer Questions

Answer each question in 1–2 sentences.

1. **Write one difference between merge() and concat().**

2. **What is data wrangling? Give one example.**

3. **What is groupby() used for?**

4. **Write one difference between a pivot table and a crosstab.**

5. **Why is a line chart a good choice for monthly sales data?**

Section 7 — Practical Programs

Write and run each program in Python. Fill the blanks (___) using the hint.

Practical 1: Join two tables (merge)

1. Create two DataFrames, customers and orders, that share a customer_id column.
2. Use merge() to join them on customer_id.
3. Print the combined DataFrame.

Hint:

```
import pandas as pd

customers = pd.DataFrame({"customer_id": [1, 2, 3],
                          "name": ["Asha", "Ravi", "Meena"]})
orders = pd.DataFrame({"customer_id": [1, 2, 2],
                       "item": ["Pen", "Book", "Bag"]})

result = pd.merge(customers, orders, on="___")
print(result)
```

Expected output: *Three rows: Asha–Pen, Ravi–Book, Ravi–Bag. (Meena has no order, so she is not shown.)*

Practical 2: Total and average sales by region (groupby)

1. Create a DataFrame with the columns region and sales.
2. Use groupby("region") and find the sum and mean of sales.
3. Print the result.

Hint:

```
import pandas as pd

df = pd.DataFrame({"region": ["North", "South", "North", "South", "East"],
                   "sales":  [100, 200, 300, 400, 500]})

result = df.groupby("___")["sales"].agg(["___", "mean"])
print(result)
```

Expected output: *East 500 / 500.0, North 400 / 200.0, South 600 / 300.0 (sum / mean).*

Practical 3: Monthly sales charts (matplotlib and Seaborn)

1. Create a DataFrame with the columns month and sales.
2. Draw a line chart of sales using matplotlib.
3. Draw a bar chart of the same data using Seaborn.

Hint:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.DataFrame({"month": ["Jan", "Feb", "Mar", "Apr"],
                   "sales": [100, 150, 130, 180]})

plt.plot(df["month"], df["___"])      # line chart
plt.title("Monthly Sales")
plt.show()

sns.barplot(x="month", y="sales", data=___)   # bar chart
plt.show()
```

Expected output: *Two charts: a line chart and a bar chart of sales for Jan to Apr.*

Instructor Answer Key

Section 2 — MCQ

Q	Answer	Why
1	B	pd.read_csv() loads a CSV file into a DataFrame.
2	C	merge() joins tables on a common column.
3	D	groupby() splits data into groups, then applies a function.
4	A	Reshaping changes the layout (wide ↔ long), e.g. with melt or pivot.
5	A	A pivot table summarizes values across two categories.
6	C	Seaborn is built on matplotlib and has nicer default styles.
7	B	plt.plot() draws a line chart.
8	D	Data ordered by time is a time series.
9	B	pd.crosstab() counts how often value pairs occur.
10	C	Fixing missing or duplicate values is a core wrangling task.

Section 3 — Fill in the Blanks

1. merge() 2. pivot 3. groupby 4. Seaborn 5. time series

Section 4 — True or False

1. True 2. False 3. True 4. False 5. True

Section 5 — Match the Following

1 → b 2 → d 3 → f 4 → e 5 → a 6 → c

Section 6 — Short Answers (Key Points)

- merge() joins tables on a common column. concat() just stacks tables one below the other (or side by side).
- Cleaning and preparing raw data for analysis. Example: fixing missing values or removing duplicates.
- It splits data into groups, applies a function (sum, mean, count) to each group, and combines the results.
- A pivot table summarizes values (sum, mean, etc.). A crosstab usually shows counts between two columns.
- A line chart shows how values change over time, so trends and ups and downs are easy to see.

Section 7 — Practical Programs (Sample Solutions)

Practical 1

```
result = pd.merge(customers, orders, on="customer_id")
print(result)
```

Practical 2

```
result = df.groupby("region")["sales"].agg(["sum", "mean"])
print(result)
```

Practical 3

```
plt.plot(df["month"], df["sales"])
plt.title("Monthly Sales")
plt.show()

sns.barplot(x="month", y="sales", data=df)
plt.show()
```